Kenny Cao

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Education

University of California, San Diego **Expected June 2029**

Bachelor of Science in Computer Science, Emphasis in Mathematics *La Jolla, CA*

– **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Databases, Operating Systems

Experience

Stealth Startup **July 2025 – Present**

Software Engineering Intern

– Engineered an agentic full-stack sales automation platform that leverages autonomous LLM agents to qualify leads and orchestrate B2B outreach workflows.
– Built a high-concurrency backend using FastAPI and LangChain, optimizing agent tool-calling efficiency by implementing a Redis-based state management system.
– Improved lead conversion rates by 35% and reduced manual data entry overhead for sales teams by 50+ hours per week through automated CRM synchronization.

University of California, San Diego **Sept. 2025 – Present**

Software Research Assistant *La Jolla, CA*

– Developed an LLM-powered narrative engine designed to generate state-aware, dynamic quest lines and NPC interactions in interactive adventure games.
– Optimized context window utilization by building a custom vector-retrieval system (RAG) in Python to inject relevant world-building data into model prompts.
– Achieved a 20% increase in response formatting accuracy and reduced token consumption by 15% through refined few-shot prompt engineering.

Projects

Distributed Real-Time Trading Engine | *Python, Redis, Docker, AWS* **January 2026**

– Architected a high-frequency trading simulation engine capable of processing 10,000+ mock transactions per second with sub-millisecond latency.
– Deployed a containerized microservices architecture using Docker and AWS ECS, utilizing Redis Pub/Sub for low-latency communication between services.

Omni-Auth: Secure Microservices Gateway | *Go, gRPC, JWT, OAuth, TLS, PostgreSQL* **Oct. 2025**

– Engineered a distributed authentication gateway for microservices to centralize identity management and session validation across a heterogeneous tech stack.
– Implemented JWT/OAuth-based stateless authentication over TLS-secured gRPC interceptors, optimizing session lookups with a write-through caching strategy.
– Reduced cross-service authentication latency by 25% and centralized security policy enforcement for over 15 distinct API endpoints.

Scale-Share: Cloud-Native Image Host | *Go, AWS S3, CloudFront, Terraform* **Aug. 2025**

– Designed a scalable image hosting service designed to handle global content delivery with high availability and automated optimization.
– Provisioned infrastructure as code using Terraform to manage AWS S3 buckets and CloudFront distributions, ensuring 99.9% uptime for global asset delivery.

Leadership

Build with Gemini × XPRIZE Hackathon **August 2026**

Technical Team Lead *La Jolla, CA*

– Led a team of three in developing an AI-powered web interface that translates web content, including menus and pricing, into a user's preferred language in real time.
– Integrated Google Gemini for real-time translation and coordinated the team from ideation through prototype development and final submission.

Technical Skills

Languages: Python, Java, C#, C, C++, JavaScript, TypeScript, Swift, Go, HTML, CSS, XML, Lua, SQL
Frameworks & Databases: React.js, Next.js, Node.js, FastAPI, Django, Ruby on Rails, Spring Boot, gRPC, PostgreSQL, MongoDB, Redis
Cloud & DevOps: AWS (S3, ECS), Azure, Docker, Terraform
Other: OAuth, SSL/TLS, Embedded Systems & Microcontrollers, Software Testing & Debugging, Git, Figma, PyTorch, TensorFlow, VS Code, GitHub, Android Studio, Linux, Cursor